Normal Distribution

Group - 4

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New Balance Case Study

"Fashions fade, style is eternal."

- Yves Saint Laurent

Data Source

kaggle

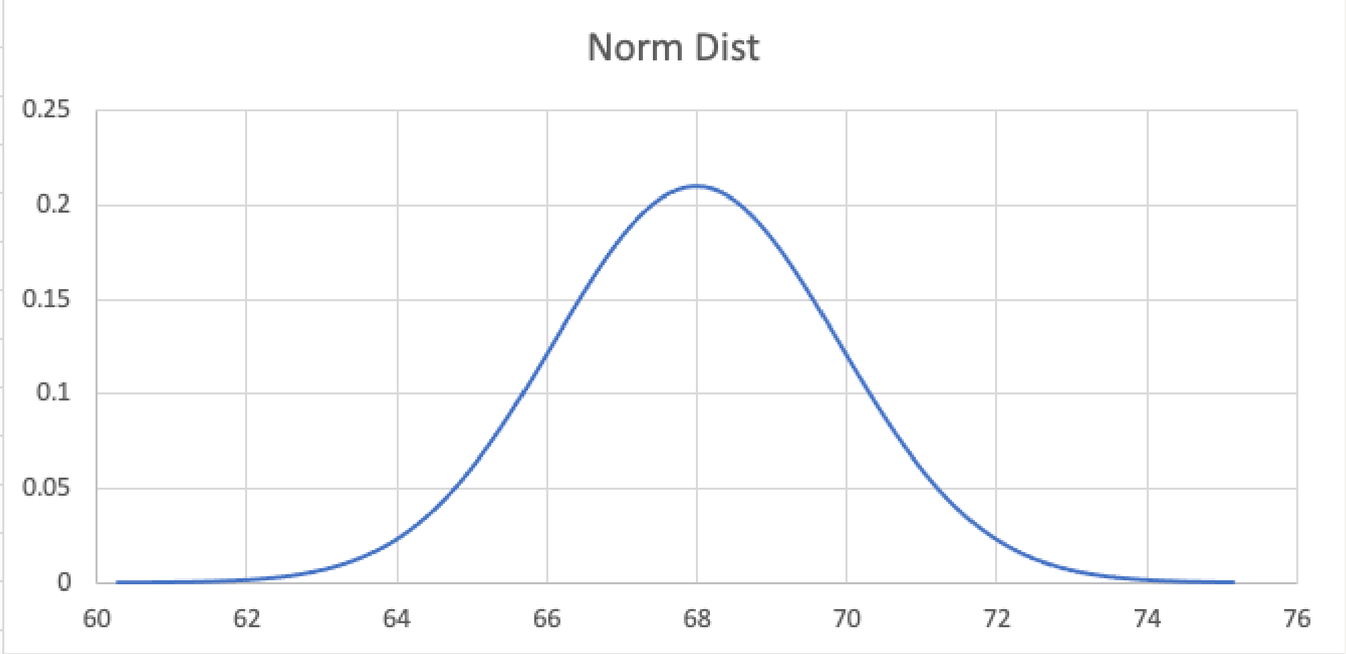
https://www.kaggle.com/code/dilipkumar2k6/normal-distribution-and-z-score/data

Challenges!!!

Analysis Objective

To determine the height distribuiton of an Indian male consumer. So, as to decide the proportional quantity to be manufactured for upcoming male sportswear line up into the Indian market.

Analysis



Mean - 68 inches

Std Dev. - 1.90 inches

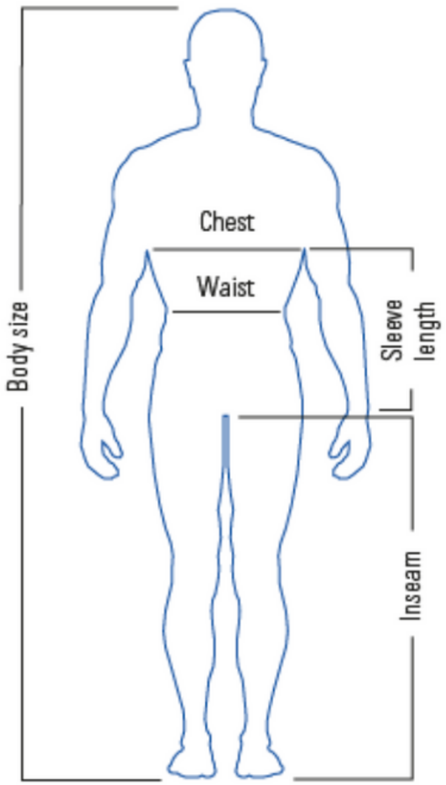
Chi-Square Test

- Chi-square is a statistical test commonly used to compare observed data with data we would expect to obtain according to a specific hypothesis.
- For example, if, according to Mendel's laws, you expected 10 of 20 offspring from a cross to be male and the actual observed number was 8 males, then you might want to know about the "goodness to fit" between the observed and expected.
- Were the deviations (differences between observed and expected) the result of chance, or were they due to other factors.
  - How much deviation can occur before you, the investigator, must conclude that something other than chance is at work, causing the observed to differ from the expected.

CONCLUSION & DECISION TAKEN

- The average height as per the data set containing 25000 samples is 68 inches (5.6 ft).
- The standard deviation is of 1.9
- The company decided to manufacture the male sportswear lineup according to Indian size.
- Where 34-34 (68)% of the total men's sportswear will be made for sizes for heights between **68-70 & 66-68 inches.**
- 13.5 -13.5 (27)% of the total men's sportswear will be made for sizes, for heights between **70 -72 & 64-66 inches.**
- 2.35 - 2.35 (4.7)% of the total men's sportswear will be made for sizes, for heights between **72-74 & 62-64 inches.**
- 0.15 - 0.15 (0.3)% of the total men's sportswear will be made for sizes, for heights between **74-76 & 60-62 inches.**

Clothing size chart – body height

Men

Man‘s sizes	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70
Body height	156	160	164	168	171	174	177	180	182	184	186	188	190	192	194	196	198
Chest measurement	76	80	84	88	92	96	100	104	108	112	116	120	124	128	132	136	140
Waist measurement	64	68	72	76	80	84	88	92	98	102	108	114	120	126	132	138	144
Leg length	71	72	73	74	76	78	79	80	81	82	83	84	84	85	86	87	88
Front sleeve length	40	41	42	43	44	45	46	47	47,5	48	48,5	49	49	49,5	50	50,5	51